

Utility Maximization 2: Extensions

Mark Dean

GR6211 - Microeconomic Analysis 1

- We have now proved the following theorem

Theorem

A Choice Correspondence on a finite X has a utility representation
if and only if *it satisfies axioms α and β*

- We have now proved the following theorem

Theorem

A Choice Correspondence on a finite X has a utility representation
if and only if *it satisfies axioms α and β*

- Great! We know how to test the model of utility maximization!
- However, our theorem is only as useful as the data set we are working with
- As discussed at the time, there are some problems with the data we have assumed so far

- What are some issues with this data set?
- ① Observe choices from all choice sets
- ② We allow for people to choose more than one option
 - i.e. we allow for data of the form

$$C(\{kitkat, jaffacakes, lays\}) = \{jaffacakes, kitkat\}$$

- ③ X Finite

- ① What if X is not Finite?
- ② What if we don't Observe Choices from all Choice Sets?
- ③ What if we don't Observe a Choice Correspondence?

- So far we have assumed that the set of available alternatives is finite

Theorem

*A Choice Correspondence on a **finite** X has a utility representation if and only if it satisfies axioms α and β*

- So far we have assumed that the set of available alternatives is finite

Theorem

*A Choice Correspondence on a **finite** X has a utility representation if and only if it satisfies axioms α and β*

- However, this may be limiting

- So far we have assumed that the set of available alternatives is finite

Theorem

*A Choice Correspondence on a **finite** X has a utility representation if and only if it satisfies axioms α and β*

- However, this may be limiting
 - Choice from lotteries
 - Choice from budget sets
- Can we drop the word 'finite' from the above theorem?

- Remember we proved the theorem in three steps
 - ① Show that if the data satisfies α and β then we can find a preference relation $\succeq$ which represents the data
 - ② Show that if $\succeq$ is a preference relation then we can find a utility function u which represents them
 - ③ Show that if the data has a utility representation then it must satisfy α and β

What if X is not Finite?

- Remember we proved the theorem in three steps
 - ① Show that if the data satisfies α and β then we can find a preference relation $\succeq$ which represents the data
 - ② Show that if $\succeq$ is a preference relation then we can find a utility function u which represents them
 - ③ Show that if the data has a utility representation then it must satisfy α and β
- Where did we make use of finiteness?

What if X is not Finite?

- In fact the problems relating choice to preference maximization are relatively minor
- The main issue here is that, if we want to define choice on **all** subsets of X we cannot guarantee that

$$C(A) = \{x \in A \mid x \succeq y \text{ for all } y \in A\}$$

is well defined

- Example?

What if X is not Finite?

- In fact the problems relating choice to preference maximization are relatively minor
- The main issue here is that, if we want to define choice on **all** subsets of X we cannot guarantee that

$$C(A) = \{x \in A \mid x \succeq y \text{ for all } y \in A\}$$

is well defined

- Example?
- But we can get round this relatively easily
 - For example by demanding that we only observe choices from **finite** subsets of X
 - Even if X itself is not finite
 - As we shall see later we may be able to do better than this

What if X is not Finite?

- What about the relationship between preference and utility?
- Here in the proof we made heavy use of finiteness
 - Induction
- Are we in trouble?

What if X is not Finite?

- What about the relationship between preference and utility?
- Here in the proof we made heavy use of finiteness
 - Induction
- Are we in trouble?
- Just because we made use of the fact that X was finite in that particular proof doesn't mean that it is necessary for the statement to be true
- Maybe we will be lucky and the statement remains true for arbitrary X

What if X is not Finite?

- What about the relationship between preference and utility?
- Here in the proof we made heavy use of finiteness
 - Induction
- Are we in trouble?
- Just because we made use of the fact that X was finite in that particular proof doesn't mean that it is necessary for the statement to be true
- Maybe we will be lucky and the statement remains true for arbitrary X
- Sadly not

- Some definitions you should know

Definition

The natural, or counting numbers, denoted by $\mathbb{N}$, are the set of numbers $\{1, 2, 3, \dots\}$

Definition

The integers, denoted by $\mathbb{Z}$, are the set of numbers $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

Definition

The rational numbers, denoted by $\mathbb{Q}$, are the set of numbers

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a \in \mathbb{Z}, b \in \mathbb{N} \right\}$$

- Some definitions you should know

Definition

The natural, or counting numbers, denoted by $\mathbb{N}$, are the set of numbers $\{1, 2, 3, \dots\}$

Definition

The integers, denoted by $\mathbb{Z}$, are the set of numbers $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

Definition

The rational numbers, denoted by $\mathbb{Q}$, are the set of numbers

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a \in \mathbb{Z}, b \in \mathbb{N} \right\}$$

Definition

A set is *countably infinite* if there is a bijection between that set and the natural numbers

- Here are some properties of $\mathbb{Q}$ and $\mathbb{R}$.
- ① $\mathbb{Q}$ is countable
 - ② $\mathbb{R}$ is uncountable
 - ③ For every $a, b \in \mathbb{R}$ such that $a < b$, there exists a $c \in \mathbb{Q}$ such that $a < c < b$ (i.e. $\mathbb{Q}$ is dense in $\mathbb{R}$)

Corollary

One cannot find a one to one function mapping $\mathbb{R}$ to $\mathbb{Q}$

Definition

Let $\succeq$ be a binary relation on $\mathbb{R} \times \{1, 2\}$ such that

$$\begin{aligned} \{a, b\} \succeq \{c, d\} \text{ iff} \\ \text{(i) } a > c \\ \text{or (ii) } a = c \text{ and } b \geq d \end{aligned}$$

You should check that you agree that $\succeq$ is a complete preference relation.

Fact

There is no utility function that rationalizes $\succeq$.

Utility Representation with Non-Finite X

- So what can we do in order to ensure that preferences have a utility representation?
- First things first: how big is the problem?
- The counter example above made use of the fact that X was uncountable
- Does this mean the problem goes away if X is **countably** finite?
- It turns out the answer is yes

Utility Representation with Countable X

Theorem

*If a relation $\succeq$ on a **countable** X is complete and transitive then there exists a utility function $u : X \rightarrow \mathbb{R}$ which represents $\succeq$, i.e.*

$$u(x) \geq u(y) \iff x \succeq y$$

Utility Representation with Uncountable X

- We know from the example of lexicographic preferences that we cannot replace 'countable' with 'any' X in the previous theorem
- In order to guarantee that we have a utility representation of a preference relation on an uncountable X we need another condition

- One technically useful, but not very intuitive condition is based on **order separability**
- This means that the preference relation $\succeq$ on a set X has embedded in it a countable subset that is **dense** according to $\succ$

Definition

A preference relation on $\succeq$ on a set X is order separable if there exists a countable sequence $e \in X^{\mathbb{N}}$ such that for every $x, y \in X$ such that $x \succ y$ there exists $e_i \in e$ such that

$$x \succeq e_i \succ y$$

- It turns out that order separability is enough to guarantee that a preference relation has a utility representation

Theorem

A preference relation on a set X has a utility representation if and only if it is order separable

Proof.

(i) By previous theorem we know that we can assign a utility representation to $\succeq$ on e

(ii) We can use this as 'scaffolding' to assign utilities



- Notice that the proof that Lex preferences don't have utility representation was basically showing that there was no countable order dense subset of the preferences

- Perhaps a more intuitive way to go is to insist that preferences are **continuous**
- Broadly speaking, this means that if we change the items a little bit the preferences also change only a little bit
- i.e. they don't 'jump'

- Perhaps a more intuitive way to go is to insist that preferences are **continuous**
- Broadly speaking, this means that if we change the items a little bit the preferences also change only a little bit
- i.e. they don't 'jump'

Definition

We say that a preference relation $\succeq$ on a **metric space** X is continuous if, for any $x, y \in X$ such that $x \succ y$, there exists an $\varepsilon > 0$ such that, for any $x' \in B(x, \varepsilon)$ and $y' \in B(y, \varepsilon)$, $x' \succ y'$

- Perhaps a more intuitive way to go is to insist that preferences are **continuous**
- Broadly speaking, this means that if we change the items a little bit the preferences also change only a little bit
- i.e. they don't 'jump'

Definition

We say that a preference relation $\succeq$ on a **metric space** X is continuous if, for any $x, y \in X$ such that $x \succ y$, there exists an $\varepsilon > 0$ such that, for any $x' \in B(x, \varepsilon)$ and $y' \in B(y, \varepsilon)$, $x' \succ y'$

- Examples of preferences that are not continuous?

- Perhaps a more intuitive way to go is to insist that preferences are **continuous**
- Broadly speaking, this means that if we change the items a little bit the preferences also change only a little bit
- i.e. they don't 'jump'

Definition

We say that a preference relation $\succeq$ on a **metric space** X is continuous if, for any $x, y \in X$ such that $x \succ y$, there exists an $\varepsilon > 0$ such that, for any $x' \in B(x, \varepsilon)$ and $y' \in B(y, \varepsilon)$, $x' \succ y'$

- Examples of preferences that are not continuous?
 - Lexicographic preferences
 - 'The price is right'

- Alternative characterizations of continuity:

Lemma

A preference relation $\succeq$ on a metric space X is continuous if and only if, for every $x, y \in X$ and sequence $\{x_n, y_n\}$ such that $x_n \rightarrow x$ and $y_n \rightarrow y$ then $x_n \succeq y_n \forall n$ implies $x \succeq y$

- i.e. the graph of $\succeq$ is closed

Lemma

A preference relation $\succeq$ on a metric space X is continuous if and only if, for any x the sets $\{y | y \succ x\}$ and $\{y | x \succ y\}$ are open

- You will prove for homework that all these definitions are equivalent

- One thing that is relatively easy to prove is that continuity of utility implies continuity of preference

Theorem

If a preference relation $\succeq$ can be represented by a continuous utility function then it is continuous

- One of the most famous theorems in mathematical social sciences is that continuity guarantees the existence of a continuous utility representation

Theorem (Debreu)

Let X be a separable metric space. $\succeq$ is a continuous preference relation on X if and only if it can be represented by a continuous utility function.

- One of the most famous theorems in mathematical social sciences is that continuity guarantees the existence of a continuous utility representation

Theorem (Debreu)

Let X be a separable metric space. $\succeq$ is a continuous preference relation on X if and only if it can be represented by a continuous utility function.

- Proving this in all its glory is beyond us, so we are going to prove something weaker

Theorem

Let X be a convex subset of $\mathbb{R}^n$ and $\succeq$ be a complete preference relation on X . If $\succeq$ is continuous, then it can be represented by a utility function.

- So now we have a method of dealing with utility and preferences in uncountable domains
- What about choice?
- Here we now have two issues
 - ① We need to guarantee that maximal elements exist in all choice sets
 - ② We would like to make sure the preferences that represent choices are continuous

- To deal with problem 1 we will restrict ourselves to **compact** subsets of X
- Notice that if we can guarantee continuous preferences then this solves the first problem
 - Continuous preferences are equivalent to continuous utility functions
 - Continuous functions on compact sets obtain their maximum

- To deal with problem 1 we will restrict ourselves to **compact** subsets of X
- Notice that if we can guarantee continuous preferences then this solves the first problem
 - Continuous preferences are equivalent to continuous utility functions
 - Continuous functions on compact sets obtain their maximum
- So how can we guarantee choice can be represented by continuous preferences?
- We would like choices to be continuous!
 - Choice sets that are 'close' to each other give rise to 'similar' choices

- How can we make this formal?

- How can we make this formal?
- We need a metric on sets!

Definition (The Hausdorff metric)

Let (X, d) be a metric space, and $cb(X)$ be the set of all closed and bounded subsets of X . We will define the metric space $(cb(X), d^h)$, where d^h is the Hausdorff metric induced by d , and is defined as follows: For any $A, B \in cb(X)$, define $\Lambda(A, B)$ as $\sup_{x \in A} d(x, B)$. Now define

$$d^h(A, B) = \max \{ \Lambda(A, B), \Lambda(B, A) \}$$

- We can use this to define a **continuous choice correspondence**

Definition

Let X be a compact metric space and Ω_X be the set of all closed subsets of X and $C : \Omega_X \rightarrow 2^X$ be a choice correspondence. If $S_m \rightarrow S$ for $S_m, S \in \Omega_X$, $x_m \in C(S_m) \forall m$ and $x_m \rightarrow x$, implies that $x \in C(S)$, then we say C is continuous.

- We can use this to define a **continuous choice correspondence**

Definition

Let X be a compact metric space and Ω_X be the set of all closed subsets of X and $C : \Omega_X \rightarrow 2^X$ be a choice correspondence. If $S_m \rightarrow S$ for $S_m, S \in \Omega_X$, $x_m \in C(S_m) \forall m$ and $x_m \rightarrow x$, implies that $x \in C(S)$, then we say C is continuous.

- It turns out that continuity, plus α and β , is enough to give us our desired results

Theorem

Let X be a compact metric space and Ω_X be the set of all closed subsets of X and $C : \Omega_X \rightarrow 2^X$ be a choice correspondence. C satisfies properties α , β and continuity if and only if there is a complete, continuous preference relation $\succeq$ on X that rationalizes C .